Evidence-backed, interactive health-insurance coverage simulator

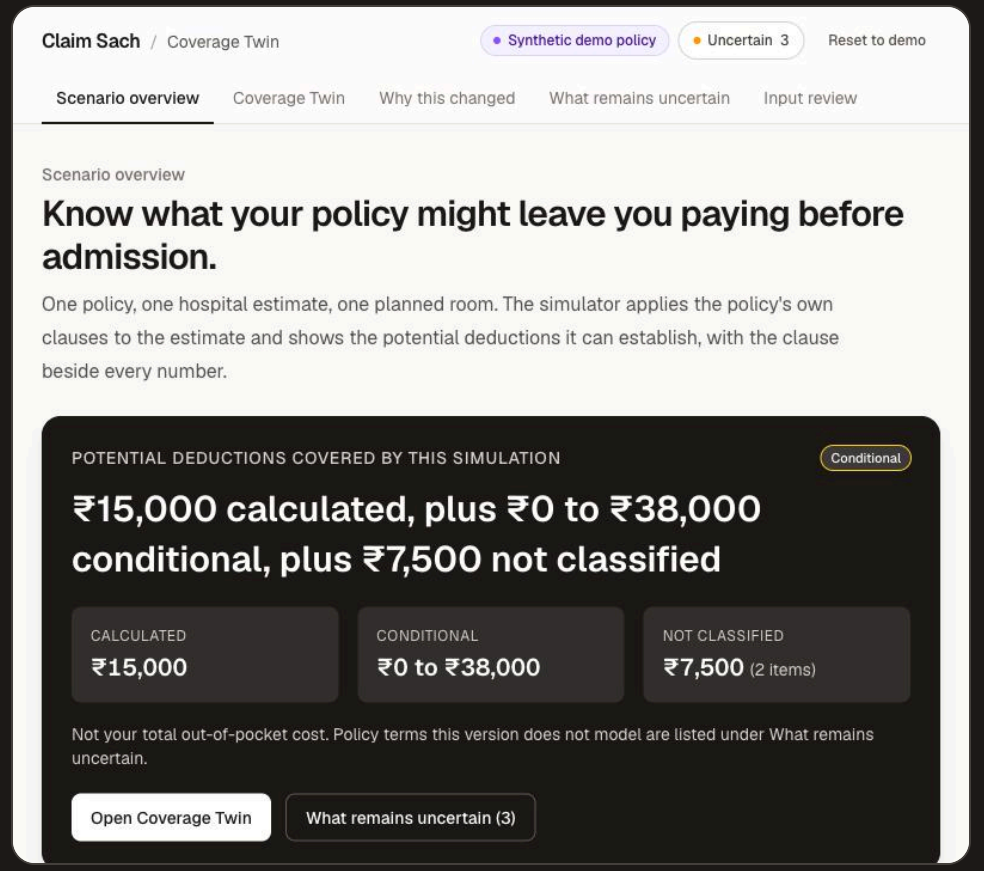
Claim Sach

Coverage Twin · Track 2: AI-Powered Financial Journeys

Know what your policy might leave you paying before admission.

Pushkar Mondal · Paytm Build for India AI Hackathon, Bengaluru · October 2026

A working prototype: deterministic engine, 180 tests, a real IRDAI-hosted wording encoded and measured, runs offline.
`github.com/Pushkarmondal/paytm-hackathon`



PROBLEM STATEMENT

The policy's cost shows up at the hospital

Room-rent limits, proportionate deductions, sub-limits and exclusions surface at admission, not at purchase.

The room-rent clause is the sharpest. Pick a room above the eligible rate and many policies pay the surgeon, anaesthetist, nursing and operation-theatre charges only in the proportion *eligible ÷ actual rate*. It is not about the room.

Which items are touched depends on the policy's own definition. General explainers stop at the concept; the gap we are testing is applying *your* wording to *your* bill.

Worked example on a synthetic policy and estimate, computed by the prototype. Not a market statistic.

₹2,500 a day

the room-rent gap in our worked example: ₹7,500 chosen against ₹5,000 eligible

₹48,000

potential deductions from that one choice, ₹38,000 of it on items that are not the room

4 of 11 lines

affected under the policy's definition; 3 excluded by it; 2 the wording never classifies

PROPOSED SOLUTION

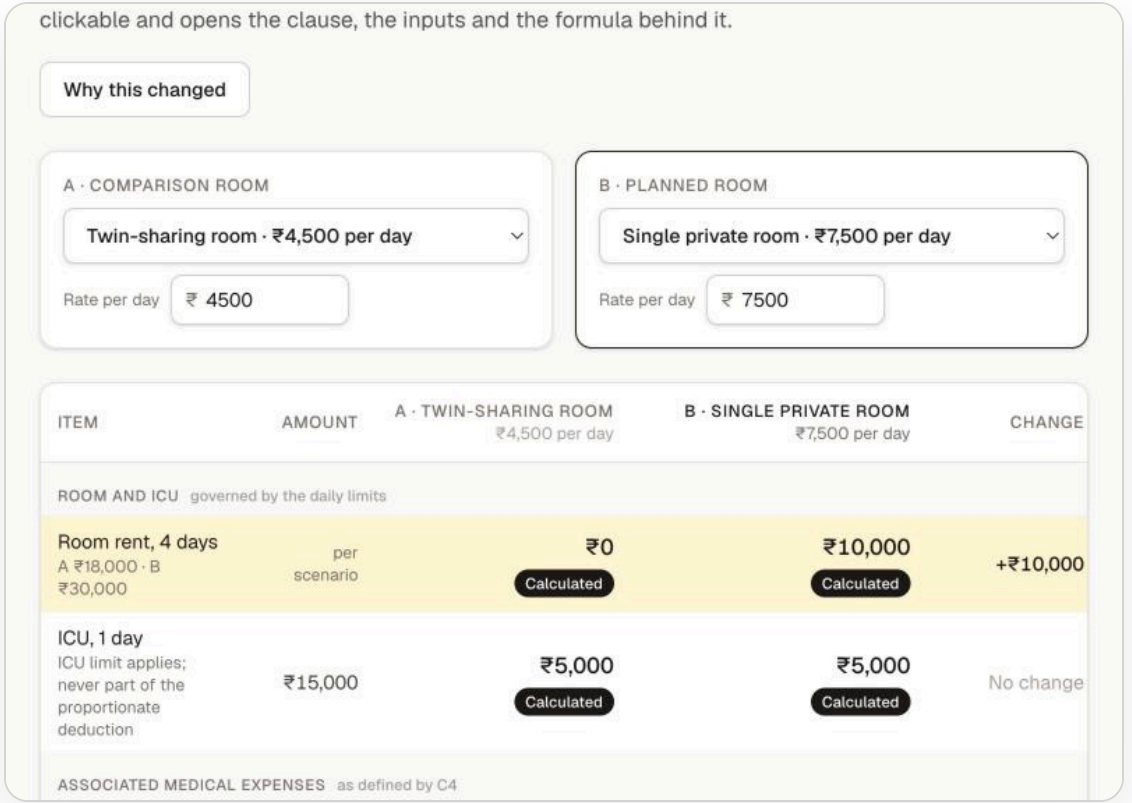
Coverage Twin: your wording, your bill, live

1 **Policy documents** | 2 **Structured rules** | 3 **Scenario simulation** | 4 **Itemised potential deductions** | 5 **Source clause** | 6 **Explicit uncertainty**

What a judge does in three minutes

- Open the example policy and estimate.
- Change the room; the itemised breakdown updates at once.
- Click any amount: clause, page, inputs with provenance, formula.
- Open the uncertainty panel: what could not be established, and who can settle it.
- Switch to a real wording and watch the same engine refuse to guess.

Not a chatbot. Not a payout guarantee. A scenario estimate of the deductions it covers, with its unknowns on screen.



TECHNOLOGY / TECH STACK

Deterministic engine, AI only with human review

Frontend

TypeScript, strict. Next.js 16 (App Router), React 19, Tailwind 4. Six screens on a tested screen model; no business logic in components.

Rules engine

Pure TypeScript, no I/O, no model. Integer paise, exact ratios, one half-up rounding. A trace on every amount: rule, inputs, clause, formula, output, open assumptions.

Two wordings

A synthetic demo policy, and Arogya Sanjeevani (New India, from irdai.gov.in): ten clauses quoted verbatim, page-cited, SHA-256 shown. Partially encoded and labelled so.

AI, in its place

The Ruleset builder: a model proposes rules with quotes; the engine verifies each quote verbatim; a person confirms. Cached in this build; the live adapter is off. Never in the calculation path.

Quality

Vitest, 180 tests, written first. GitHub Actions on every PR. Evaluation harness: six measures over two wordings, results committed and checked for staleness.

Runs anywhere

One command, offline on localhost; fonts bundled; Vercel-ready. Nothing the user enters leaves the browser. No key, no account, no analytics, no database.

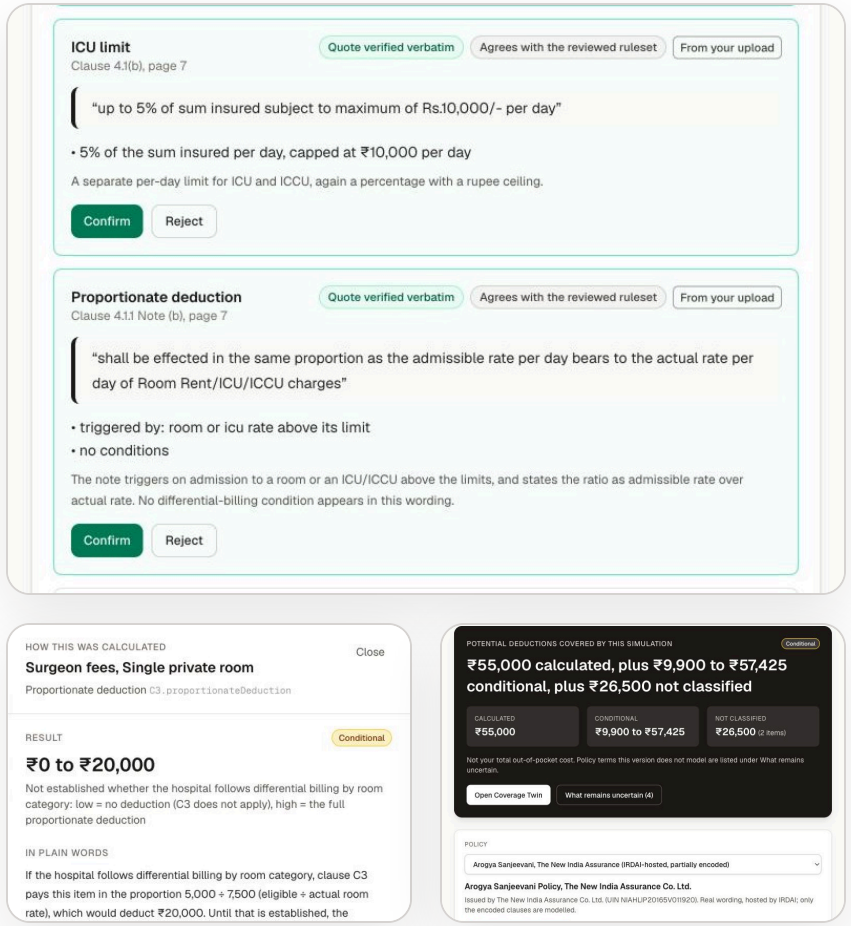
DATA FLOW, ALL IN THE BROWSER

```
fixture or upload (JSON, CSV) → ingestion: Zod schema + verbatim evidence check → screen model → rules engine: classify → limits → proportion → co-payment → traces → six screens, question sheet, evidence pack
```

AI seam, off by default: public clause text → model proposals → verbatim quote check → human confirmation → ruleset. No schedule, bill or name ever leaves the browser.

USP

Item-level, clause-cited, honest about unknowns



1. **Every number is clickable**

   Clause and page, inputs with where they came from, the formula, the operation. Tampered traces fail validation.

2. **Unknown is never zero**

   Items the wording does not classify stay out of every total. A conditional amount shows both bounds and the condition.

3. **Change your room, see the difference**

   Two rooms side by side, the rule behind every changed row; the room ladder and the question sheet turn it into a decision and a call script.

4. **Proven on a real wording, honestly**

   Arogya Sanjeevani, ten clauses verbatim with a checksum; where the 2019-20 text conflicts with IRDAI's 2020 norms, the engine says so.

Bounded review of nine public sources: none did this. We do not claim nobody does; we claim we looked.

IMPACT & BENEFITS

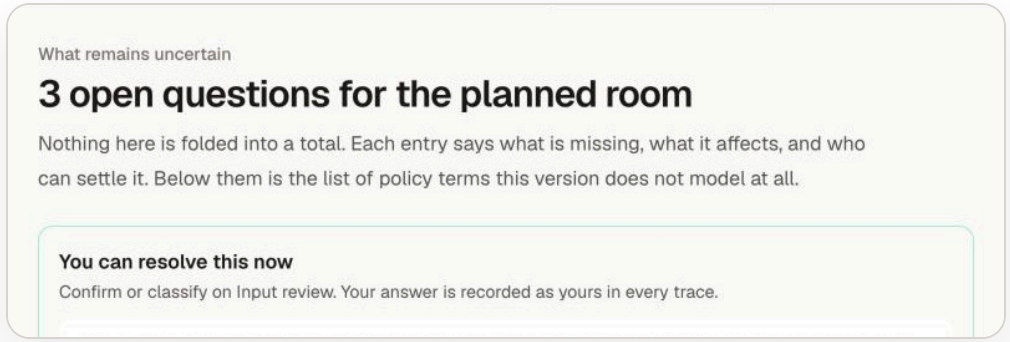
The room decision, made before admission

For the policyholder

The room decision with its deductions visible before admission, not after. Questions for the hospital and the insurer written out: what to ask, and what it is worth.

For a broker's claim-assistance agent

The clause on screen during the call. The uncertainty list is the call script; the evidence pack is the written answer.



MEASURED, NOT CLAIMED

MEASURE	SYNTHETIC	REAL WORDING
Clause retrieval, top-1	11 of 12	10 of 12
Rule extraction, field-level	10 of 10	10 of 12
Calculations, exact paise	9 of 9 · 46 amounts	3 of 3 · 21 amounts
Uncertainty and refusal	6 of 6	3 of 3
Model proposals: quotes verbatim (cached)	n/a	6 of 6
Model proposals: agree with reviewer (cached)	n/a	11 of 13

Measured on the synthetic wording and on ten encoded clauses of one real wording; the model proposals are cached, not a live run. The product shows no accuracy figure anywhere. Figures on this slide are potential deductions covered by the simulation, never total out-of-pocket cost.

BUSINESS MODEL

Where it fits Paytm, and what must be tested

VERIFIED FACTS

- Paytm Insurance Broking Pvt Ltd is a Direct (Life & General) broker: IRDAI Reg. No. 700, renewed 9 Feb 2026, valid to 16 Feb 2029.
- It distributes car, two-wheeler, health, life and term policies from 18+ insurers, including Niva Bupa and Aditya Birla Health. It does not underwrite.
- It already offers claim assistance by phone. Arogya Sanjeevani is IRDAI's standard product on one wording; one verified encoding serves every insurer that sells it.
- FY25 turnover ₹37.1 crore, loss ₹13.2 crore (Form AOC-1), inside a ₹2,593 crore financial-services line in FY26. Insurance is a small line for Paytm today; clarity is a wedge worth testing.

HYPOTHESES, EACH WITH ITS TEST

HYPOTHESIS	TEST
Fewer claim-time support contacts and disputes	Pilot cohort against a matched cohort
More eligible-room choices when the consequence is visible	Room-change rate in the simulator, then claims data
Panel insurers co-fund a coverage-clarity tool	Three conversations; one signed pilot letter is the pass
A coverage check lifts conversion or renewal on the policy page	A/B test, if Paytm chooses to run one

Revenue options depend on those tests: a broker-side feature (retention), an insurer-paid integration, or a premium claim-assistance tier. No Paytm integration is claimed or built. Bima Sugam is not live; it belongs in the future tense.

Built, tested, runs offline

`github.com/Pushkarmondal/paytm-hackathon · npm install && npm run dev`

180 unit tests · two wordings, one of them real · six evaluation measures · a ruleset builder where a model proposes and a person confirms · a trace on every number.

Next: a second real wording through the builder with the live adapter on; independent review of the evaluation fixtures; a four-week pilot at a broker's claim-assistance desk.

